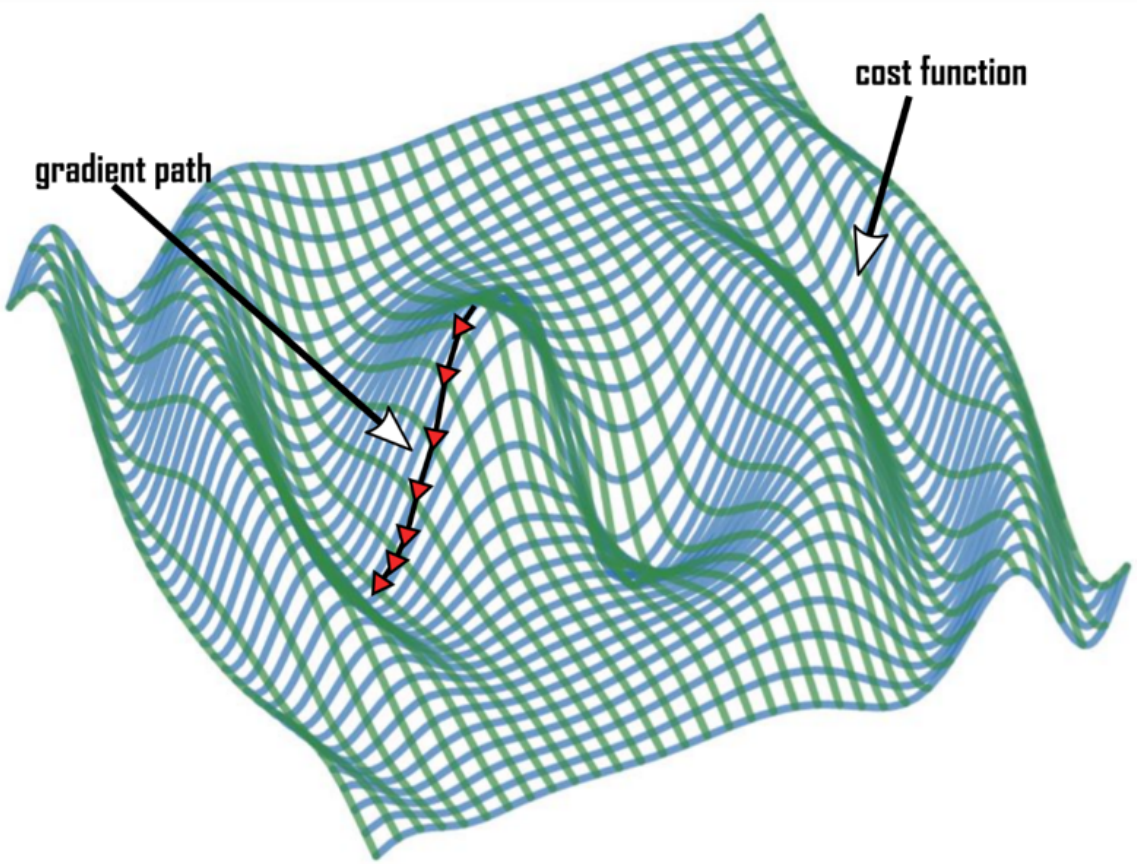


Figure 16.15 Gradient descent applied to a function of 2 variables. The cost function defines a surface.

As for the starting point `P0`, you might wonder how do we choose it: there are different ways, but – unless you have domain knowledge telling you otherwise – it's best to choose it randomly, and possibly run the optimization several times, starting each time from a new randomly chosen point, and keeping track of the best overall result.

Can you see it? We are back to random sampling, only applying a sophisticated local optimization heuristic after each sample is taken.

16.3.3 When is Gradient Descent Appliable?